

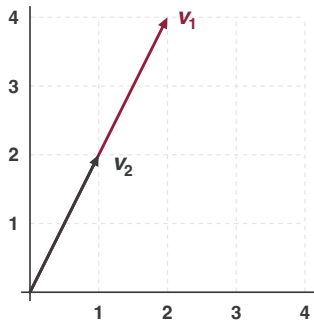
Rank and the Determinant

ECON 441: Introduction to Mathematical Economics
Lecture 6

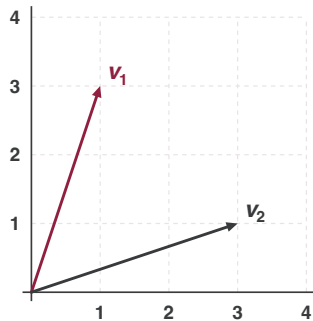
Div Bhagia

Linear Independence

Linearly dependent



Linearly independent



Rank of a Matrix

Rank of a matrix = maximum number of linearly independent rows

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Rank of A ? Rank of B ?

Full rank $\iff$ Nonsingularity

Checking for Nonsingularity

Rank of a matrix = maximum number of linearly independent rows or (equivalently) columns

If a square matrix has full rank, it is nonsingular.

To check for nonsingularity, calculate the **determinant**: for singular matrices, the determinant is zero.

Determinant

Determinant $|A|$ is a unique scalar associated with a **square** matrix A .

Determinant of a 2×2 Matrix:

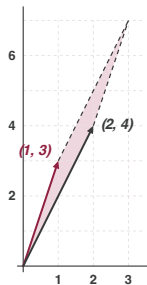
$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Can be calculated as:

$$|A| = a_{11}a_{22} - a_{12}a_{21}$$

Determinant: Geometric Interpretation

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$



Determinant of a 3×3 Matrix

$$\begin{aligned} |A| &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \\ &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \end{aligned}$$

Determinant of a $n \times n$ Matrix

A **minor** of the element a_{ij} , denoted by $|M_{ij}|$ is obtained by deleting the i th row and j th column.

Cofactor C_{ij} is defined as:

$$|C_{ij}| = (-1)^{i+j} |M_{ij}|$$

Then,

$$|A| = \sum_{i=1}^n a_{ij} |C_{ij}| = \sum_{j=1}^n a_{ij} |C_{ij}|$$

Homework Questions

- Exercise 5.2: 1 (c), (e), (f), 2, 3, 6

Textbook reference: 5.1, 5.2